

Predictive maintenance 4.0 driving efficiency and sustainability in industry

Unlocking the full potential of IoT and AI technologies for proactive maintenance, optimising schedules, reducing downtime, and extending asset lifespan, to achieve peak efficiency, sustainability, and safety in industrial operations.

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maintenance has become a driving force in the industry, offering significant cost savings. With the power of IoT and AI technologies, companies can unlock the full potential of predictive maintenance, ensuring peak efficiency, sustainability, and safety in their industrial operations.

Why is predictive maintenance 4.0 driving the industry?

In the era of the fourth revolution, predictive maintenance 4.0 takes a proactive approach to maintenance by analysing data in real-time to identify potential issues before they cause any significant downtime. Predictive maintenance has become a promising approach and driving force in the industry due to its ability to optimise maintenance schedules, reduce downtime, increase asset lifespan, leading to a significant reduction in maintenance cost. An industry report by McKinsey states that there is a potential for some companies to improve equipment availability by 5-15 per cent and reduce maintenance costs by 18-25 per cent by adopting fully digitised predictive maintenance. This clearly means that the adoption of predictive maintenance 4.0 is expected to continue to grow, providing companies a competitive advantage in an increasingly competitive marketplace.

Downtimes are expensive, especially when they're unplanned. Unplanned machine downtime can contribute up to 24 per cent of costs to modern industrial production while causing damage to reputation, efficiency, the environment, and the bottom line. With most maintenance work being categorised as 'crisis work' to fix breakdowns, the losses quickly add up through labour, raw materials, and lost potential product sold.

To ensure sustainability and safety in industrial operations, production assets should be operating at peak efficiency, without wasting energy, damaging the environment, or enlarging your carbon footprint.

Predictive Maintenance 4.0 is gaining importance to avoid the above losses and become competitive along with ensuring sustainability and safety in the industrial operations.

What are some of the key benefits of implementing predictive maintenance in an Industry 4.0 environment?

The primary impact of unplanned downtime halts the production and costs industrial manufacturers about \$50 billion a year. A study by Deloitte also states that a non-optimized maintenance strategy reduces the production capacity by 5 to 20 per cent. Predictive maintenance, by utilising a stream of IoT sensors, collects and shares data that enables companies to predict failures over time, reduce unnecessary inspections and repairs and

How does Industry 4.0 technology, such as the Internet of Things (IoT) and artificial intelligence (AI), facilitate predictive maintenance?

Advanced Industry 4.0 technologies such as Artificial intelligence (AI), Machine Learning (ML), the Internet of Things (IoT), Cloud Computing, and Big Data have transformed equipment maintenance and product line management approach of the businesses. IoT Sensors collect enormous quantities of data and translate it into meaningful insights. Further, with the help of AI algorithms, this recorded data is then processed and analysed to detect anomalies, identify patterns, predict future maintenance needs, and recommend the best course of action.

By continuously tracking the performance of equipment, manufacturers may anticipate maintenance requirements before they arise, preventing expensive downtime and cutting maintenance expenses.

Often confused with preventative maintenance, predictive maintenance begins to make use of industrial IoT capabilities to identify more precisely when equipment requires maintenance – as close to failure as possible – to get maximum uptime and reduce maintenance costs. Predictive maintenance uses data gathered from IoT-connected equipment continuously over time and provides a far more precise trend profile based on performance. Data is collected as the equipment is running, so it doesn't need to be taken offline.

With data from sensors monitoring equipment and performing automated data analytics governed by an IoT-based predictive maintenance solution, companies eliminate the guesswork that characterises scheduled maintenance and instead can leverage insights based on real-time measurements, reducing and often eliminating errors.

What are some of the challenges that organisations may face when implementing predictive maintenance in an Industry 4.0 context?

Even though adopting predictive maintenance in an industrial context is inevitable, it is surrounded by challenges that hinder the application and collective adoption of this smart maintenance approach. Firstly, the technology, data aggregation, data science expertise, and new processes to turn monitoring into predicting can be costly. Furthermore, shifting from one technology to another, requirement of highly skilled and trained professionals can be a time-consuming process, so it is also a big challenge. Lastly, it is necessary to ensure the security of asset data and information – which is a concern that mostly exists on the end user's side.

How do you see predictive maintenance evolving in the next few years? What new technologies or approaches are emerging that could have an impact?

In the next five years of time span, predictive maintenance will have undergone a revolution due to the pressing need to extend the lifespan of ageing industrial machinery.. A study suggests that the global predictive maintenance market is anticipated to reach US \$19.24 billion by 2028. Key trends include the widespread use of AI and ML allowing more accurate and timely maintenance, IoT sensors such as smart metres, the use of inspection technologies, and digital twins, allowing businesses to monitor their assets and visualise future scenarios.

Predictive maintenance will mature into Prescriptive maintenance in the future. While predictive maintenance enables smarter and faster root-cause analysis, reduces unnecessary downtime and provides visibility into the health of remote machines, prescriptive maintenance moves facilities to a more automated approach. With an IoT-based prescriptive maintenance approach, industrial facilities gain the ability to have the maintenance system resolve issues autonomously.

With IoT, companies can further augment the power of prescriptive maintenance using AI and machine learning in combination with sensors to diagnose the root cause of problems, indicate appropriate remedial actions and manage the entire maintenance process. An IoT-based prescriptive system also allows companies to automate all aspects of maintenance, including ordering the required parts, scheduling the service, accounting for the time and cost, keeping track of parts on hand and ensuring that the job is informed. All these steps can be performed by the system autonomously in a fraction of the time required by any previous maintenance scheme.

How can organisations measure the success of their predictive maintenance program? What key performance indicators (KPIs) should they track?

Manufacturers typically look to improve equipment availability, enhance productivity, and reduce maintenance costs through a predictive maintenance program. It would help to assess the current (as-is) status and define the future (to-be) status upfront so as to be clear on the anticipated ROI on a predictive maintenance implementation.

Predictive maintenance is already being deployed by companies that want to elevate their maintenance programs to the highest levels of maturity. For example, a service provider that helps rail operators increase their return-on-assets (ROA) while improving the reliability and safety of rail transport. The company's goal is to help organisations reduce maintenance costs and unplanned downtime, as well as increase the efficiency of operation planning while reducing risks

The solution also increased the efficiency of operation planning, which reduced maintenance costs and energy consumption, making its rail service more competitive with other transport options. Because the system also performs much of the maintenance it identifies, the company reduces unnecessary transfers to maintenance, leading to greater operational efficiency.

As a result, the firm:

- Proved asset reliability of greater than 99 percent
- Optimised operation planning with up to 20 percent fewer delays
- Achieved more effective root-cause analysis and reduced complex fault resolution times by more than 20 percent

How important is data analytics in predictive maintenance? What role does it play in ensuring the accuracy and effectiveness of the approach?

Data analytics plays a critical role in predictive maintenance in the manufacturing industry. By collecting and analysing vast amounts of data generated by machinery, manufacturers can identify patterns and trends that may indicate potential equipment failures before they occur.

With the help of machine learning algorithms, data analytics can accurately predict equipment failures and prescribe maintenance actions that can save manufacturers time and money. It further helps manufacturers improve the reliability and efficiency of their equipment, ensuring they operate at peak performance while reducing maintenance and repair costs. Needless to say, Big data and predictive analytics are transforming predictive maintenance, and by default, helping to transform Industry 4.0 organisations.

Finally, what are some of the ethical considerations that organisations should keep in mind when implementing predictive maintenance in an Industry 4.0 context?

Firstly, businesses must adhere to the applicable policy laws and regulation when gathering and analysing data. Further, accuracy of the data collected is of utmost importance as it could result in costly repairs and legal ramifications. Finally, besides taking appropriate security measures to prevent data breaches, the results of predictive maintenance must be monitored to guarantee that predictive maintenance is done properly and sustainably.

